

strchr in C

Last Updated : 23 Jul, 2025

The `strchr()` function in C is a predefined function in the `<string.h>` library. It is used to find the first occurrence of a character in a string. It checks whether the given character is present in the given string. If the character is found, it returns the pointer to its first occurrence otherwise, it returns a null pointer, indicating the character is not found in the string.

Syntax of `strchr()` in C

```
char *strchr(const char *str, int ch);
```

Parameters of `strchr()` in C

str: It is the string in which we have to search the character. This string is a constant character pointer, meaning that the function will not modify the string.

ch: It is a character to be searched in the string. Though passed as an int, it represents a character and is cast internally. This allows `strchr()` to be used with character values, including special characters and ASCII values.

Return Value of `strchr()`

The `strchr()` function returns a pointer to the first occurrence of the character in the string. If the character is not found, the function returns `NULL`. Here, the return type is `char *`, which allows direct access to the character and subsequent characters in the string from the found position.

Example Programs Using `strchr()` in C

The following examples illustrates how we can use `strchr()` in various scenarios.

Example 1:

Using `strchr()` to check the existence of character in a string and print its first occurrence.

```
#include <stdio.h>
```

```
#include <string.h>
```

```
â\200\213
```

```
int main()
```

```
{
```

```
    // define a string
```

```
    const char* str = "GeeksforGeeks";
```

```
    // define a char ch to be searched in str
```

```
    char ch = 's';
```

```
â\200\213
```

```
    // Use strchr to find the first occurrence of the
```

```
    // character 's'
```

```
    const char* result = strchr(str, ch);
```

```
â\200\213
```

```
    if (result != NULL) {
```

```
        // Calculate the position by subtracting the base
```

```
        // pointer from the result pointer
```

```
        printf("Character '%c' found at position: %ld\n",
```

```
            ch, result - str);
```

```
    }  
    else {  
        printf("Character '%c' not found.\n", ch);  
    }  
    â\200\213  
    return 0;  
}
```

Output

Character 's' found at position: 4